

PROJECT REPORT
ON
AI FRAUD DETECTION IN CRYPTO USING MACHINE LEARNING
RESEARCH PAPER
22CS422
BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE AND ENGINEERING



Submitted by:

Harsh Wadhwa

2210990382

Supervised By:

Dr. Shikha Tuteja

Associate Professor

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CHITKARA UNIVERSITY

CHANDIGARH-PATIALA NATIONAL HIGHWAY, RAJPURA, PUNJAB, INDIA

CERTIFICATE

This is to certify that the project report titled “**AI Fraud Detection in Cryptocurrency Using Machine Learning**” submitted by **Harsh Wadhwa** is a bonafide work carried out under the guidance and supervision of **Dr. Shikha Tuteja**.

The work presented in this report is original and has not been submitted elsewhere for any degree or academic certification.

Harsh Wadhwa
2210990382

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my project guide, faculty members, and the Department of Computer Science Engineering for their support and guidance throughout the completion of this project.

I also thank my friends and classmates who helped me during the development and documentation process. Finally, I am grateful to online research communities, open-source contributors, and publicly available datasets that made this project possible.

CONTENTS

S No.	Title	Page No.
1.	Abstract	5
2.	Introduction	6
3.	Methodology	7
4.	Tools and Technology	8
5.	Implementation	9
6.	Major Findings/Outcomes/Results	10
7.	Conclusion	11
8.	References/Appendices	12

ABSTRACT

Cryptocurrency fraud has become a major concern due to the rapid growth of blockchain technology and decentralized financial systems. Fraudulent activities such as phishing attacks, fake investment schemes, money laundering, wallet theft, and pump-and-dump operations continue to increase across digital financial platforms. Traditional fraud detection systems based on static rules are often unable to detect modern and evolving fraud patterns effectively.

This project presents a machine learning-based framework for detecting fraudulent cryptocurrency transactions using blockchain transaction analytics. The proposed framework integrates preprocessing techniques, feature engineering methods, and multiple machine learning classification algorithms including Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, and XGBoost.

The system was evaluated using cryptocurrency transaction datasets obtained from blockchain repositories and Kaggle datasets. Experimental analysis showed that XGBoost and Random Forest achieved the best overall fraud detection performance due to their ability to handle complex behavioural relationships within transaction data.

The project also discusses ethical concerns, privacy issues, operational challenges, and future research directions related to AI-driven blockchain fraud detection systems.

Introduction

Cryptocurrency technology has transformed digital financial systems by introducing decentralized transactions without requiring central authorities such as banks. Blockchain networks provide transparency, immutability, and fast global transactions. However, the rapid expansion of cryptocurrency systems has also increased fraudulent activities within digital financial ecosystems.

Fraudulent activities such as phishing scams, Ponzi schemes, fake exchanges, wallet theft, and money laundering continue to threaten users and financial platforms. Traditional fraud detection systems mainly rely on manually designed rules and predefined thresholds. These systems often fail to detect newly emerging fraud patterns effectively.

Machine learning provides a more intelligent approach for fraud detection by analysing transaction behaviour and identifying suspicious patterns automatically. Machine learning models can study transaction timing, transfer frequency, wallet connectivity, and behavioural indicators to classify fraudulent activities.

This project focuses on developing a machine learning-based cryptocurrency fraud detection framework capable of identifying suspicious blockchain transactions using multiple supervised learning algorithms.

Methodology

The development of the proposed cryptocurrency fraud detection system follows a structured machine learning workflow designed for efficient fraud identification and blockchain transaction analysis. The methodology includes multiple stages such as data collection, preprocessing, feature engineering, model training, evaluation, and fraud prediction.

Requirement Analysis

The initial phase focused on identifying the major system requirements. The framework required secure handling of blockchain transaction data, accurate fraud classification, scalable processing capability, and reliable evaluation metrics.

The main objectives included:

- Detecting fraudulent cryptocurrency transactions
- Analysing blockchain transaction behaviour
- Comparing machine learning model performance
- Generating fraud alerts automatically

System Design

The system was designed using a layered machine learning architecture. The architecture contains data preprocessing, feature extraction, classification, evaluation, and fraud monitoring components.

The framework supports both graph-based blockchain transaction datasets and tabular cryptocurrency datasets.

Machine Learning Implementation

Multiple machine learning algorithms were implemented and compared, including:

- Logistic Regression
- Decision Tree
- Random Forest
- Support Vector Machine
- XGBoost

Each model was trained using identical experimental conditions to ensure fair evaluation.

Tools and Technologies

Programming Language

- Python – Used for machine learning model development and blockchain transaction analysis.

Machine Learning Libraries

- Pandas – Used for data preprocessing and manipulation.
- NumPy – Used for numerical computation.
- Scikit-learn – Used for machine learning algorithms and evaluation metrics.
- XGBoost – Used for gradient boosting classification.
- Matplotlib – Used for graph generation and result visualization.

Development Tools

- Visual Studio Code – Used for coding and debugging.
- Git & GitHub – Used for version control and project hosting.

Dataset Sources

- Kaggle Cryptocurrency Fraud Dataset
- Elliptic Bitcoin Transaction Dataset

Implementation

The implementation of the cryptocurrency fraud detection framework was carried out by integrating preprocessing techniques, feature engineering methods, machine learning algorithms, and evaluation modules into a complete fraud analysis system.

Data Handling

Cryptocurrency transaction datasets were imported and processed using Python libraries. Data cleaning operations were performed to improve dataset consistency and remove noisy records.

Feature Engineering

Important transaction features such as transaction frequency, transfer amount, wallet connectivity, and transaction timing were extracted from blockchain datasets.

Machine Learning Training

Multiple machine learning models were trained independently using the prepared datasets. Performance comparison was conducted using standard fraud detection evaluation metrics.

Result Visualization

Graphs and comparative performance charts were generated for all models to analyse prediction performance visually.

Major Findings / Outcomes / Results

The experimental analysis demonstrated that machine learning algorithms can effectively identify fraudulent cryptocurrency transactions.

Major findings include:

- XGBoost achieved the highest overall fraud detection performance.
- Random Forest performed strongly on graph-based blockchain datasets.
- Ensemble learning techniques produced better results than traditional linear models.
- Recall proved to be one of the most important metrics because missed fraud cases may result in financial losses.
- Dataset structure significantly influenced machine learning performance.

The project successfully demonstrated the practical use of machine learning for blockchain security applications.

CONCLUSION

The project successfully developed a machine learning-based cryptocurrency fraud detection framework capable of analysing blockchain transaction behaviour and identifying suspicious activities automatically.

Multiple supervised learning algorithms including Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, and XGBoost were evaluated using cryptocurrency transaction datasets. Experimental results showed that ensemble learning models such as XGBoost and Random Forest achieved superior fraud detection performance due to their ability to capture complex transaction relationships.

The project also highlighted important challenges related to scalability, privacy, fairness, and ethical AI deployment in blockchain security systems.

Overall, the framework demonstrates the potential of machine learning and blockchain analytics in improving digital financial security and fraud prevention systems.

References

1. S. Nakamoto, “Bitcoin: A Peer-to-Peer Electronic Cash System,” 2008.
2. T. Chen and C. Guestrin, “XGBoost: A Scalable Tree Boosting System,” ACM SIGKDD, 2016.
3. Scikit-learn Documentation. Available at: <https://scikit-learn.org>
4. Kaggle Cryptocurrency Fraud Dataset. Available at: <https://www.kaggle.com>
5. Elliptic Bitcoin Dataset Documentation.
6. Pandas Documentation. Available at: <https://pandas.pydata.org>
7. NumPy Documentation. Available at: <https://numpy.org>
8. Matplotlib Documentation. Available at: <https://matplotlib.org>

Appendices

Appendix A: System Architecture

This appendix contains the layered architecture of the proposed cryptocurrency fraud detection framework, including preprocessing, feature extraction, classification, and fraud alert modules.

Appendix B: Dataset Information

This section contains details about the cryptocurrency transaction datasets used during experimentation.

Appendix C: Performance Graphs

This appendix contains individual model comparison graphs and combined evaluation charts generated during experimental analysis.